EnterpriseRAG-Bench · PILOT

Benchmark report · 4Minds · EnterpriseRAG-Bench d36685e, official metrics-based evaluation

Benchmark: onyx-dot-app/EnterpriseRAG-Bench on GitHub

Abstract

Does 4Minds answer questions about a company's own documents better than the RAG systems that customers already use? 8 systems with the same model (GPT-OSS-120B) answered the same 50 questions from the same 10,000 company documents. The official EnterpriseRAG-Bench scorer, with its own judge (gpt-5.4), scored each system alone.

Official scorer, unchanged · *Default settings* · *Leaderboard format*

PILOT: small sample and a corpus slice. Do not use for conclusions and do not compare with the public leaderboard.

Headline

1 Overall score

The leaderboard score (0–100): for each question, the completeness of the answer when the judge says that the answer is correct, else 0. Bars show the mean; the white whisker is the 95% bootstrap confidence interval. You cannot directly compare these scores with the public leaderboard results, because this pilot uses only a slice of the corpus (10,000 of 511,958 documents).

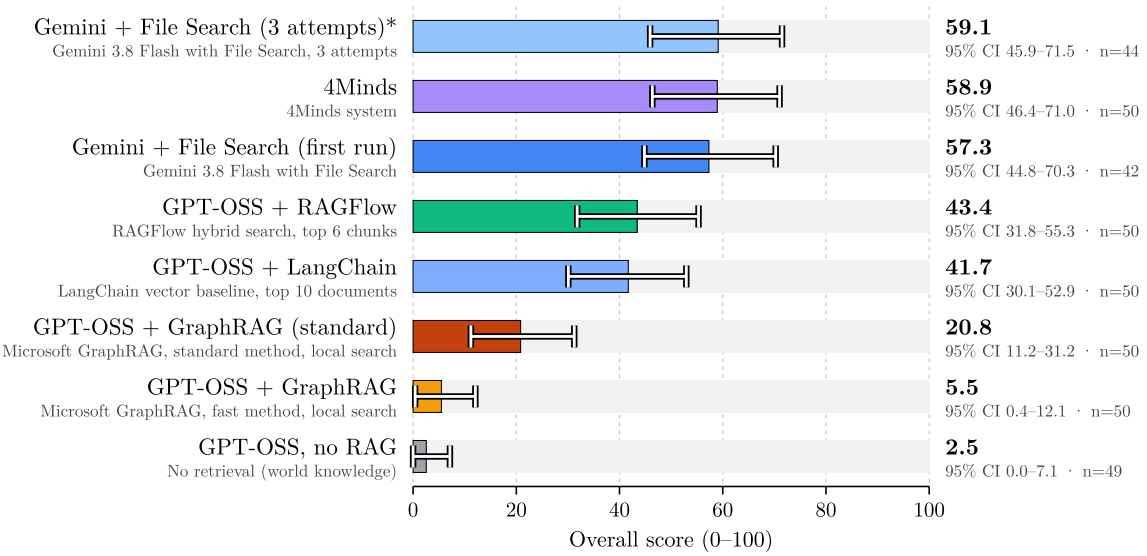


Figure 1: Overall score

All metrics

2 Official metrics

From the official `results.json` of each system. Recall and invalid extra documents use the document IDs that the system gave; they apply only to questions with gold documents. "Overall, missing = 0" counts a question that the official scorer skipped (no answer) as 0.

Table 1: Official metrics

System	Overall	Overall, missing = 0	Correctness	Completeness	Recall	Invalid extra docs	Corrected	Scored
Gemini + File Search (3 attempts)*	**59.11**	52.02	81.82	68.07	90.41	0.10	1	44/50
4Minds	**58.93**	58.93	70.00	66.24	80.50	11.93	2	50/50
Gemini + File Search (first run)	**57.33**	48.15	80.95	66.71	89.93	0.10	1	42/50
GPT-OSS + RAGFlow	**43.41**	43.41	74.00	48.93	83.15	3.70	1	50/50
GPT-OSS + LangChain	**41.69**	41.69	76.00	45.88	82.13	8.76	1	50/50
GPT-OSS + GraphRAG (standard)	**20.84**	20.84	38.00	28.27	35.08	5.30	2	50/50
GPT-OSS + GraphRAG	**5.49**	5.49	14.00	14.67	3.20	4.61	2	50/50
GPT-OSS, no RAG	**2.55**	2.50	8.16	9.22	0.00	0.00	0	49/50

Corrected: questions where 3 judges agreed that a retrieved document that is not gold has the answer, so that document became gold for that system (official default).

Not scored

3 Systems and questions without a result

When a system did not give an answer to a question, the official scorer skipped that question. The official averages do not include these questions. The column "Overall, missing = 0" in the table above counts each of them as 0.

- **Gemini + File Search (first run)**: 8 of 50 questions skipped. For each of these questions, Gemini made searches for 90 to 120 seconds, and then the Gemini API stopped the request with the error "Model generated too many tool calls".
- **Gemini + File Search (3 attempts)***: 6 of 50 questions skipped. Each of these questions got 3 attempts, and each attempt stopped with the same "too many tool calls" error.
- **GPT-OSS, no RAG**: 1 of 50 questions skipped. This system gets no documents, and for this question the model gave an empty answer because it had no documents to answer from.

Quality and speed

4 Overall score and latency

One point for each system. Move the pointer over a point for the 95% confidence interval, the 90th-percentile latency and the cost.

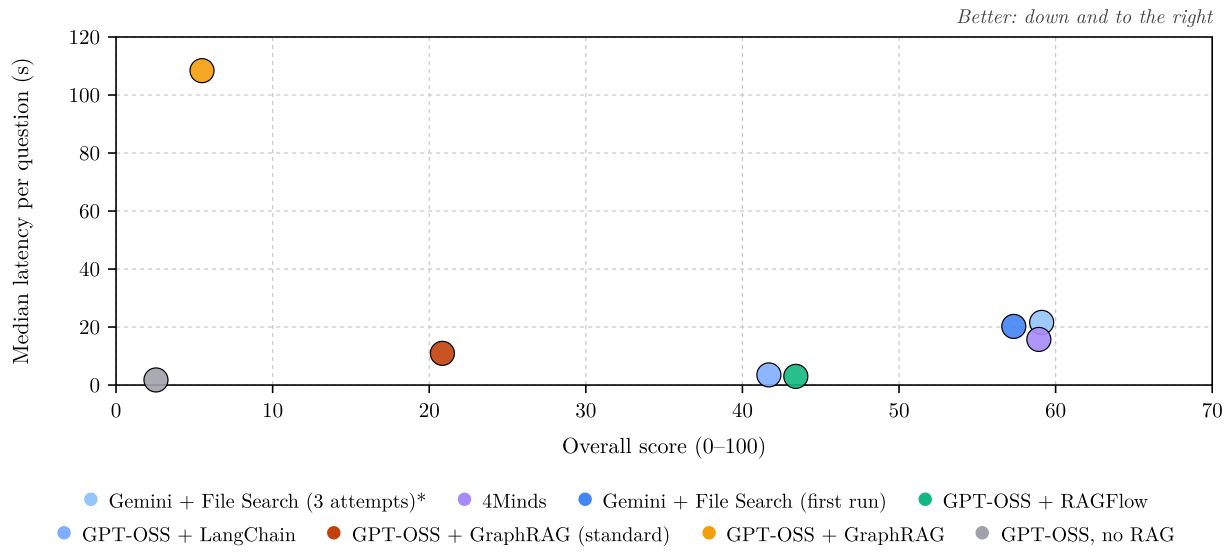


Figure 2: Overall score and latency

Table 2: Overall score, latency, cost and documents per answer for each system (the point details of Figure 2)

System	Overall score	95% CI	Latency, median	Latency, p90	Est. answer cost	Documents per answer
Gemini + File Search (3 attempts)*	**59.1**	45.9–71.5	21.61 s	76.58 s	$4.81	1.36
4Minds	**58.9**	46.4–71.0	15.73 s	33.01 s	n/a (subscription)	13.0
Gemini + File Search (first run)	**57.3**	44.8–70.3	20.22 s	76.58 s	$4.34	1.36
GPT-OSS + RAGFlow	**43.4**	31.8–55.3	3.0 s	8.98 s	$0.05	4.98
GPT-OSS + LangChain	**41.7**	30.1–52.9	3.5 s	12.99 s	$0.13	10.0
GPT-OSS + GraphRAG (standard)	**20.8**	11.2–31.2	10.94 s	21.37 s	$0.10	5.9
GPT-OSS + GraphRAG	**5.5**	0.4–12.1	108.39 s	126.09 s	$0.10	4.74
GPT-OSS, no RAG	**2.5**	0.0–7.1	1.76 s	4.65 s	$0.01	0.0

Latency depends on the number of workers, the network and the load on each service during the run.

By question type

5 Overall score per question type

Each type has only 2 to 17 questions in this pilot, so one question changes a score by many points.

Table 3: Overall score per question type

System	basic	complete-ness	conflicting info	con-strained	high level	info not found	intra document reasoning	miscel-laneous	project related	semantic
Gemini + File Search (3 attempts)*	62.9 n=14	33.3 n=2	30.0 n=2	97.8 n=3	0.0 n=2	–	75.0 n=4	20.0 n=2	45.9 n=4	70.7 n=11
4Minds	69.4 n=17	29.2 n=2	50.0 n=2	100.0 n=3	0.0 n=2	100.0 n=2	83.3 n=4	40.0 n=2	38.0 n=4	45.2 n=12
Gemini + File Search (first run)	62.9 n=14	33.3 n=2	30.0 n=2	100.0 n=2	0.0 n=2	–	66.7 n=3	20.0 n=2	45.9 n=4	70.7 n=11
GPT-OSS + RAGFlow	28.3 n=17	29.2 n=2	73.3 n=2	84.4 n=3	0.0 n=2	100.0 n=2	29.2 n=4	30.0 n=2	35.9 n=4	59.2 n=12
GPT-OSS + LangChain	30.6 n=17	22.5 n=2	50.0 n=2	91.4 n=3	0.0 n=2	100.0 n=2	29.2 n=4	30.0 n=2	50.7 n=4	47.1 n=12
GPT-OSS + GraphRAG (standard)	24.1 n=17	8.3 n=2	37.5 n=2	93.3 n=3	0.0 n=2	50.0 n=2	25.0 n=4	0.0 n=2	15.1 n=4	0.0 n=12
GPT-OSS + GraphRAG	1.2 n=17	0.0 n=2	0.0 n=2	0.0 n=3	0.0 n=2	100.0 n=2	0.0 n=4	0.0 n=2	13.6 n=4	0.0 n=12
GPT-OSS, no RAG	0.0 n=17	0.0 n=1	0.0 n=2	0.0 n=3	0.0 n=2	50.0 n=2	0.0 n=4	0.0 n=2	0.0 n=4	2.1 n=12

Method

6 How the benchmark ran

6.1 Protocol

- Dataset: EnterpriseRAG-Bench (Onyx), commit d36685e27371, 511,958 documents and 500 questions.
- Pilot: 50 questions, stratified by question type, seed 42.
- Corpus slice: the 74 gold documents of the pilot questions plus 9,926 random other documents (48,872,100 characters).
- Every system got the same documents and the instructions of the official answer prompt.
- Scoring: the official metrics_based_eval.py, unchanged, with its default settings and judge (gpt-5.4; gpt-5-mini for JSON recovery).
- No system answered a question before its ingestion was complete (a readiness check gates each run).

6.2 Systems

All "GPT-OSS" systems use GPT-OSS-120B on Amazon Bedrock and the official answer prompt. The GPT-OSS systems with retrieval also use text-embedding-3-large embeddings.

- 4Minds: 4Minds full end-to-end system. Leaderboard name: "4Minds system".
- GPT-OSS + LangChain: LangChain plain vector search. The model gets the 10 best full documents. Leaderboard name: "LangChain vector (text-embedding-3-large) + GPT-OSS-120B".

  Exact settings. TokenTextSplitter (512 tokens, 10% overlap), FAISS; 100 chunks to the top 10 full documents.
- GPT-OSS + GraphRAG: Microsoft GraphRAG 3.3.0, fast method. A low-cost graph made from noun phrases. Leaderboard name: "Microsoft GraphRAG 3.3.0 fast, local search (text-embedding-3-large) + GPT-OSS-120B".

  Exact settings. Fast indexing method with default settings (1,200-token chunks, noun-phrase graph, community reports by GPT-OSS-120B); local search context (community level 2).
- GPT-OSS + GraphRAG (standard): Microsoft GraphRAG 3.3.0, standard method. A knowledge graph that the LLM builds. Leaderboard name: "Microsoft GraphRAG 3.3.0 standard, local search (text-embedding-3-large) + GPT-OSS-120B".

Exact settings. Standard indexing method with default settings (1,200-token chunks, LLM entity and relationship extraction, description summaries and community reports by GPT-OSS-120B); local search context (community level 2).

- GPT-OSS + RAGFlow: RAGFlow v0.27.2 (self-hosted). Keyword and vector search together, with default settings. Leaderboard name: "RAGFlow v0.27.2 hybrid search (text-embedding-3-large) + GPT-OSS-120B".

  Exact settings. Naive chunking (512 tokens, DeepDOC); POST /api/v1/retrieval with the default chat-assistant settings (6 chunks, hybrid keyword + vector search with vector weight 0.3, similarity threshold 0.2, no reranker).

- Gemini + File Search (first run): gemini-3.8-flash with Google's hosted File Search tool, with default settings. Leaderboard name: "Gemini File Search (gemini-3.8-flash), first run".

  Exact settings. Gemini Developer API (Interactions API); top_k 5 per search; Gemini default temperature and thinking level; one store file per document with the File Search default chunking and embedding; the official instructions as the system instruction.

- Gemini + File Search (3 attempts)*: The first run of Gemini + File Search, with up to two more attempts for the questions that failed. Leaderboard name: "Gemini File Search (gemini-3.8-flash), 3 attempts".

  Exact settings. Same settings as the first run; the third attempt with a 300 s client timeout.

- GPT-OSS, no RAG: an empty Context Documents section, so the model uses only its own knowledge (world-knowledge baseline). Leaderboard name: "GPT-OSS-120B, no retrieval".

Terminology

7 EnterpriseRAG-Bench · metrics-based

Official metrics_based_eval.py at commit d36685e, unchanged, default settings. Judge gpt-5.4 (gpt-5-mini only for JSON recovery). Each system is scored alone; the leaderboard Space's transform_raw_data.py makes the rows.

Table 4: Metrics

Metric	Used on	Why	How
Overall score	All questions (headline, leaderboard column)	One number that needs a correct answer and rewards a complete one. The public leaderboard sorts by it.	Per question: completeness if the answer is correct, else 0. Mean over the scored questions. The official scorer skips a question with no answer and no documents; the report also shows the score with such a question as 0.
Correctness	All questions	Does the answer agree with the gold answer as a whole (no wrong or conflicting claim)?	One judge verdict (true or false) per answer against the question and the gold answer. A judge failure gives false. A question without a gold answer counts as correct.
Completeness	All questions	How much of the needed information the answer has.	The judge checks each gold fact alone against the answer. Score = facts found / all facts. No facts = 100; a judge failure on any fact gives 0.
Recall	Questions with gold documents	Did the system find the documents that have the answer? Needs the document IDs that the system used.	Gold documents in the system's document list / all gold documents.
Invalid extra docs	Questions with gold documents	Shows a system that returns many documents to raise its recall. Lower is better; it is a count, not 0–100.	Number of returned documents that are not gold and not marked "valid" (relevant) for the question.
Document correction	Before scoring (official default)	The gold lists are not perfect. A document that a system found can have the answer too.	Three judges read each returned document that is not gold. When all agree that it has the answer, it becomes gold for that system and the gold answer can get new facts. The report counts the corrected questions.